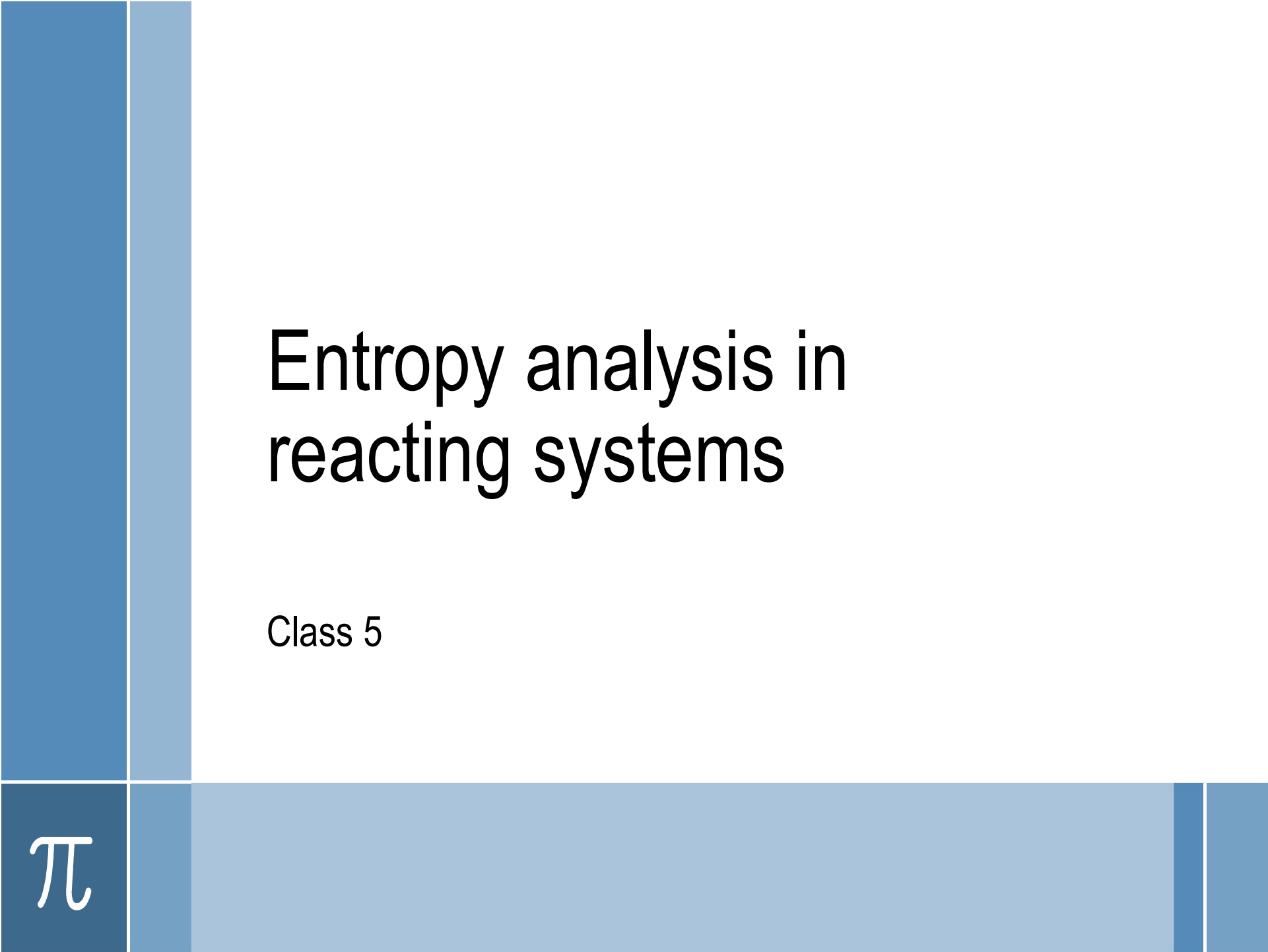


Entropy analysis in reacting systems

Class 5



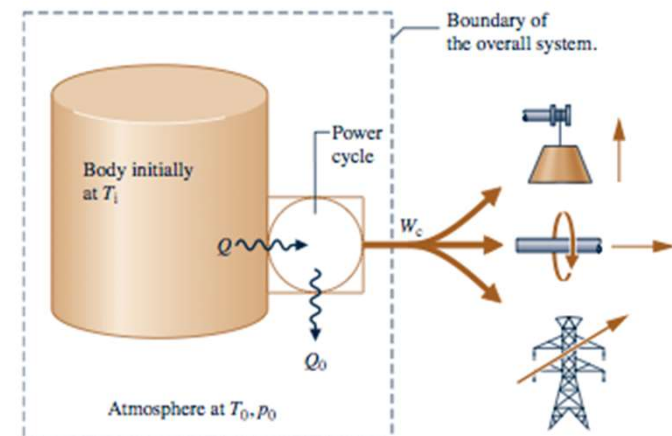
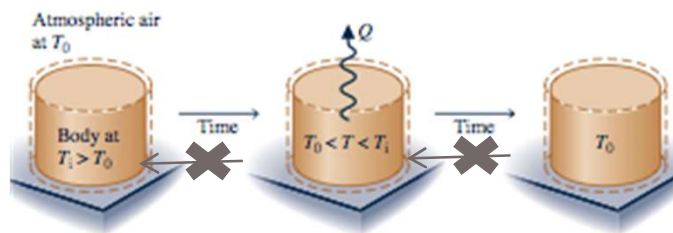
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The 2nd law of thermodynamics

- › Spontaneous processes
 - happen in one direction until equilibrium is achieved:
 - › Internally and/or with the surroundings
 - The reversible process can take place if auxiliary devices are used
- › When an imbalance exists between two systems in contact:
 - Work can be developed from them
 - There is a maximum quantity of work that can be obtained



2nd law: Clausius inequality and entropy

› Entropy (S- kJ/kmol·K):

- Energy lost in the form of heat at the boundary system during the process.
 - › Property based on Clausius inequality
 - › Disorder or uncertainty of the system
- Generated by the presence of irreversibilities
 - › Diminishing the amount of energy the process can convert/ use to develop work
- Can be accounted by an entropy balance:

$$\oint \left(\frac{\delta Q}{T} \right)_b \leq 0$$

Clausius' inequality

$$\frac{\text{Change in the amount of entropy contained within the system during a time interval}}{\text{interval}} = \frac{\text{Net amount of entropy transferred across the system boundary during the time interval}}{\text{interval}} - \frac{\text{Amount of entropy produced within the system during the time interval}}{\text{time interval}}$$

› Entropy statement of the second law:

- “It is impossible for any system to operate in a way that entropy is destroyed”
 - › There also exist the Clausius and Kelvin-Planck statements
 - Heat cannot be continuously converted into power

The 2nd law: irreversibilities

› Irreversible process

- The system and all parts of the surroundings cannot be exactly restored to their initial states after the process takes place
 - › Even if the system can be restored, the surroundings would not be able to return

› Irreversibilities:

- Heat transfer through a finite temperature difference
- Unrestrained expansion of a gas or liquid to a lower pressure
- Spontaneous chemical reaction
- Spontaneous mixing of matter at different composition states
- Friction –sliding friction, flow fluid friction
- Inelastic deformation



2nd law and entropy: applications

- › Entropy is used to:
 - Predict the direction of processes
 - Determine the best theoretical performance of cycles and devices
 - Evaluate quantitatively the factors that prevent our systems to attain the best theoretical performance
 - Establish conditions for equilibrium
- › It can also be applied outside the engineering spectrum:
 - Such as philosophy and economics

Entropy balance for non-reactive system

- › Entropy balance for a general system is expressed:

$$\underbrace{S_2 - S_1 = \Delta S_{syst}}_{\substack{\text{Net entropy change in the} \\ \text{contained amount within the} \\ \text{system}}} = \underbrace{S_{in} - S_{out}}_{\substack{\text{Net amount of entropy} \\ \text{transferred across the} \\ \text{boundary}}} + \underbrace{S_{gen}}_{\substack{\text{Generated/produced} \\ \text{amount of entropy} \\ \text{within the system}}}$$

- Entropy can be produced due to irreversibilities
- Entropy can be transferred through: mass and heat transfer

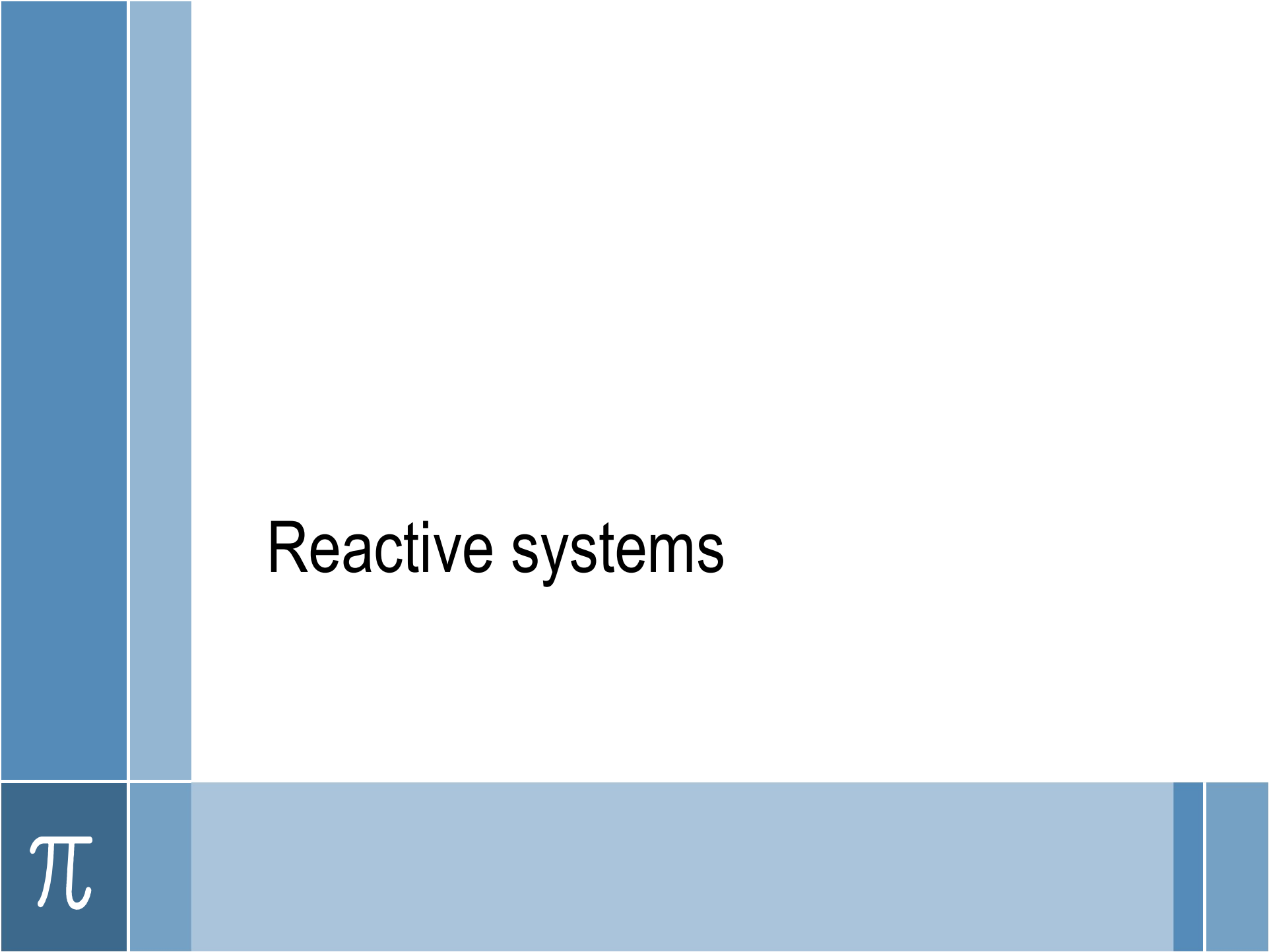
$$\Delta S_{syst} = S_Q + S_{mass} + S_{gen}$$

- Entropy accompanying heat transfer (S_Q)
 - Sign convention applies
 - b – boundary
 - › Temperature and heat at the boundary

$$S_Q = \int_1^2 \left(\frac{\delta Q}{T} \right)_b$$

- Entropy accompanying mass flow (S_{mass})
 - m_i amount of mass entering/leaving
 - Velocity and elevation have no effect on entropy

$$S_m = m_{in} s_{in} - m_{out} s_{out}$$

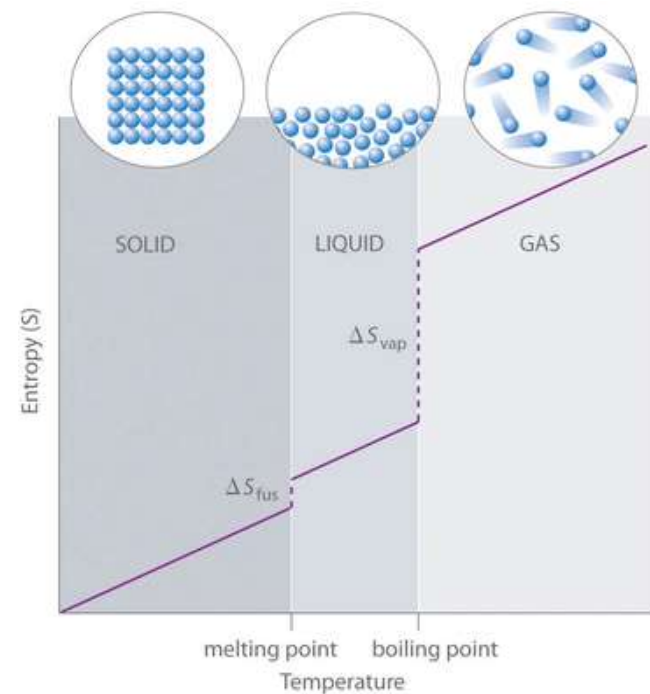


Reactive systems

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3rd Law and absolute entropy

- › To calculate entropy we require a reference state
 - An absolute base to measure the entropy of each substance involved.
- › At absolute temperature of zero (0 K) the entropy of a pure crystalline substance will be zero – 3rd law of thermodynamics.
 - Pure substance: not a mixture
 - › Component (H_2O) and elements (N)
 - › Mixtures – $S > 0$
- › Absolute entropy ($\bar{S}$)
 - Values of entropy obtained based on the datum of $S(0^\circ\text{K}) = 0 \text{ KJ/kmol}\cdot\text{K}$



Change of entropy

- › If you wanted to obtain the entropy of any pure substance at different state you would need

$$\bar{s}(T, p_{ref}) = \bar{s}^{\circ}(T_{0K}, p) + \int_{T_{0K}}^{T_M} \frac{\bar{Cp}_s}{T} dT + \frac{\Delta \bar{h}_{fs}}{T_M} + \int_{T_M}^{T_V} \frac{\bar{Cp}_f}{T} dT + \frac{\Delta \bar{h}_{fg}}{T_V} + \int_{T_V}^T \frac{\bar{Cp}_g}{T} dT$$

C_p – specific heat capacities of solid, liquid and vapour phases

h_{fg} , h_{fs} – enthalpy changes melting and vaporization at 1atm

T_V and T_M – melting and vaporization temperatures

- › Nomenclature for the tabulated data of absolute entropy
 - Absolute entropy at standard state ($\bar{s}^{\circ}$)
 - › Data of absolute entropy at reference temperature and pressure
 - 298 °K and 1atm
 - Absolute entropy at different state $\bar{s}^{\circ}(T)$
 - › Absolute entropy at certain T and pressure of 1atm (standard state)
 - Absolute entropy at different state $\bar{s}(T, p)$
 - › Absolute entropy at certain T and pressure



Tables with entropy data

ABSOLUTE ENTROPY AT 298 K AND 1 ATM

Sustancia	Fórmula	$\bar{s}^\circ$ kJ/kmol · K
Acetileno	$C_2H_2(g)$	200.85
Agua	$H_2O(l)$	69.92
Alcohol etílico	$C_2H_5OH(g)$	282.59
Alcohol etílico	$C_2H_5OH(l)$	160.70
Alcohol metílico	$CH_3OH(g)$	239.70
Alcohol metílico	$CH_3OH(l)$	126.80
Amoniaco	$NH_3(g)$	192.33
Benceno	$C_6H_6(g)$	269.20
<i>n</i> -butano	$C_4H_{10}(g)$	310.12
Carbón	$C(s)$	5.74
Dióxido de carbono	$CO_2(g)$	213.80
<i>n</i> -dodecano	$C_{12}H_{26}(g)$	622.83
Etano	$C_2H_6(g)$	229.49
Etileno	$C_2H_4(g)$	219.83
Hidrógeno	$H_2(g)$	130.68

ABSOLUTE ENTROPY FOR NITROGEN (N_2), 1 ATM AND DIFFERENT T.

T K	$\bar{s}^\circ$ kJ/kmol · K
0	0
220	182.639
230	183.938
240	185.180
250	186.370
260	187.514
270	188.614
280	189.673
290	190.695
298	191.502
300	191.682
310	192.638
320	193.562
330	194.459
340	195.328
350	196.173

Change of entropy

- › Entropy at any other state can be calculated based on the tabulated data
 - › Certain temperature at constant pressure
 - › Certain pressure at constant temperature
 - Require to know the different values
 - This can be done for compound at any state.
- › Change of entropy for an ideal gas:
- › If the analysis is of an ideal gas mixture:
 - each component has to be evaluated at mix temperature and partial pressure p_i :

$$\bar{s}(T, p) = \bar{s}(T, p_{ref}) + [\bar{s}(T, p) - \bar{s}(T, p_{ref})]$$

$$\bar{s}(T, p) = \bar{s}^\circ(T) - R \ln \frac{p}{p_{ref}}$$


 $\bar{s}_i^\circ(T) = \bar{s}^\circ + \int_{T^\circ}^T \frac{Cp_i}{T} dT$

$$\bar{s}_{mix}^\circ = \sum_i N_i \bar{s}_i^\circ$$


 $\bar{s}_i^\circ = \bar{s}^\circ(T) - \bar{R} \ln \frac{p_i y_i}{p_{ref}}$

Entropy balance for control volume at steady state

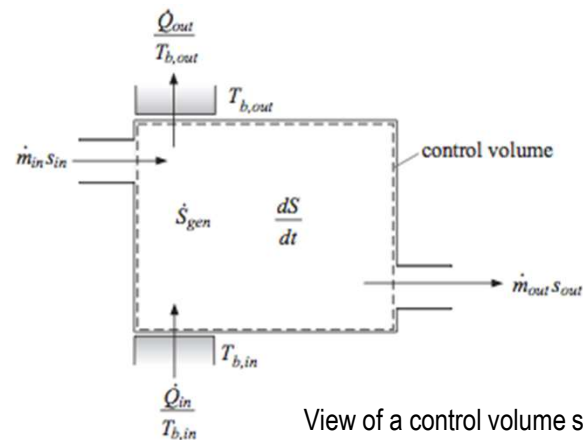
› For reacting systems:

– Balance for two-inlet, single exit reactor per mole of fuel basis:

$$0 = \sum_i \frac{q_b}{T_b} + \sum_i (n_i \bar{s}_i)_{react} - \sum_i (n_i \bar{s}_i)_{prod} + S_{gen}$$

› All entropy terms are absolute entropies

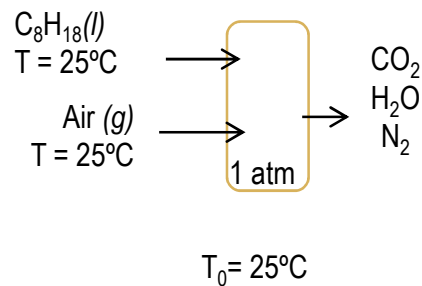
› Reacting systems combustion air and products are mixtures



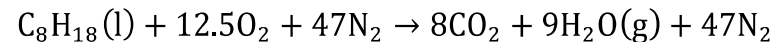
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› Liquid octane at 25 °C, 1 atm enters a well-insulated reactor and reacts with air entering the same temperature and pressure. The products of combustion exist at 1 atm pressure. For steady-state operation and negligible effects of kinetic and potential energy, determine the rate of entropy production, in kJ/kmol·K of fuel, complete combustion with the theoretical amount air.

– Assumptions: 1) all gases behave as ideal gases, 2) $E_K = E_P = 0$, 3) $W = 0$, 4) $Q = 0$, 5) Air is represented by 79% N_2 and 21% O_2



Balanced equation representing the reaction:



The problem requires to find the adiabatic temperature for this system: for which ever method Using the most basic one, we have:

$$T_{\text{prod}} = T_{\text{ad}} = 2\,395\text{K}$$

Following the entropy balance for a control system, that experiences no heat transfer:

$$0 = \bar{s}_{C_8H_{18}} + (12.5\bar{s}_{O_2} + 47\bar{s}_{N_2}) - (8\bar{s}_{CO_2} + 9\bar{s}_{H_2O} + 47\bar{s}_{N_2}) + S_{\text{gen}}$$

Using the following equation, the first for the liquid compound and the second for the mixture

$$\bar{s}(T, p) = \bar{s}(T, p_{\text{ref}}) + [\bar{s}(T, p) - \bar{s}(T, p_{\text{ref}})]$$

$$\bar{s}_i^\circ = \bar{s}^\circ(T) - \bar{R} \ln \frac{p_i y_i}{p_{\text{ref}}}$$

The entropy of the reactants is: 12
178.41 kJ/kmol_{fuel} · °K

		N_i	y_i	$s^\circ(T)$	$R_u \ln(y_i P_m)/P^\circ$	$N_i s_i^\circ$
Fuel	$C_8H_{18}(l)$	1	1	360.79	---	360.79
Air	$O_2(g)$	12.5	0.21	205.04	-12.975	2 725
	$N_2(g)$	47	0.79	191.61	-1.959	9092.62
		59.5	1			12 178.41

$s(298\text{K}, 1\text{atm})$

Using the following equation for the mixture gases

$$\bar{s}_i^\circ = \bar{s}^\circ(T) - \bar{R} \ln \frac{p_i y_i}{p_{ref}}$$

	N_i	y_i	$s^\circ(T)$	$R_u \ln(y_i P_m / P^\circ)$	$N_i s_i^\circ$
CO ₂ (g)	8	0.125	320.173	-17.2885	2 699.69
H ₂ O (g)	9	0.1406	273.986	-16.311	2 612.673
N ₂ (g)	47	0.7344	258.503	-2.567	12 270.29
	64	1			17 582.653

s (2 395K, 1atm)

The entropy of the reactants is: 17 582.653 kJ/kmol_{fuel} · °K

$$S_{gen} = (17 582.653) - 12 178.41$$

The entropy generation is: 5 404 kJ/kmol_{fuel} · °K

Energy that is being lost due to irreversibilities

Entropy balance for closed systems

- › Entropy balance for a process of a closed system which undergoes a chemical reaction:

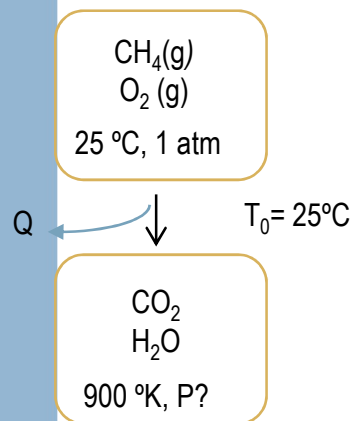
$$S_{prod} - S_{react} = \int \frac{Q_b}{T_b} + S_{gen}$$

- › S_{react} and S_{prod} represent the entropy of the reactants and products
 - In this scenario:
 - › The fuel and oxidizer are not entering as separate inlets

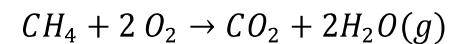
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- > A mixture of 1 kmol of gaseous methane and 2 kmol of oxygen initially at 25 °C and 1 atm burns completely in a closed, rigid container. Heat transfer occurs until the products are cooled to 900 K. If the reactants and products each form ideal gas mixtures determine the amount of heat transfer, the final pressure and the change in entropy of the system.

– Assumptions: 1) KE=PE= 0, 2) W=0, 3) all gases behave as ideal gases, 3) Air is represented by 79% N₂ and 21% O₂.



Balanced equation representing the reaction:



The problem requires to find the transferred heat:

$$Q = -745\,436 \text{ kJ/kmol}_{\text{fuel}}$$

And the final pressure of the system:

$$P_2 = P_{\text{prod}} = 3.02 \text{ atm}$$

Following the entropy balance described for a closed system:

$$S_{\text{prod}} - S_{\text{react}} = \int \frac{Q_b}{T_b} + S_{\text{gen}} \longrightarrow \Delta S_{\text{syst}} = S_{\text{prod}} - S_{\text{react}}$$

Using the following equation the gas mixture

$$\bar{s}_i^\circ = \bar{s}^\circ(T) - \bar{R} \ln \frac{p_i y_i}{p_{\text{ref}}}$$

The entropy of the reactants is:
612.096 kJ/kmol_{fuel} · °K

	N _i	y _i	s°(T)	R _u ln(y _i P _m)/P°)	N _i s _i °
CH ₄ (g)	1	1/3	186.16	-9.1338	195.294
O ₂ (g)	2	2/3	205.04	-3.371	416.822
	3	1			612.096

	N_i	y_i	$s^\circ(T)$	$R_u \ln(y_i P_m / P^\circ)$	$N_i s_i^\circ$
$\text{CO}_2(g)$	1	1/3	263.559	0.0552	263.504
$\text{H}_2\text{O}(g)$	2	2/3	228.32	5.818	444.98
	3	1			708.598

$s(900\text{K}, 1\text{atm})$
 $1\text{atm} \rightarrow 3.02\text{atm}$

The entropy of the reactants is: $708.598 \text{ kJ/kmol}_{\text{fuel}} \cdot ^\circ\text{K}$

$$S_{\text{gen}} = (708.598) - 612.096$$

The entropy generation is: $96.414 \text{ kJ/kmol}_{\text{fuel}} \cdot ^\circ\text{K}$
 Energy that is being lost to irreversibilities

$$\Delta S_{\text{syst}} = S_{\text{gen}} - \left(-\frac{Q}{T_0} \right) = 96.414 + \frac{745\,436}{298} = 2597.87 \frac{\text{kJ}}{\text{kmol}_{\text{fuel}} \text{K}}$$